MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS)
FIFTH SEMESTER EXAMINATION
Discipline Specific Elective (DSE) –MG5DSECHE302 - ENVIRONMENTAL CHEMISTRY
MODEL QUESTION PAPER
(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

1. Define environmental pollution. [K] / [CO1]
2. Explain the impact of toxic chemicals on enzymes. [U] / [CO1]
3. What are the biochemical effects of carbon monoxide (CO). [U] / [CO1]
4. How will you measure the alkalinity of water. [K] / [CO2]
5. How would you determine the presence of fluoride in water sample? [U] / [CO2]
6. Name the methods used for the determination of Lead and Mercury in water. [K] / [CO2]
7. Describe the method used for the analysis of SO_2 in air . [U] / [CO3]
8. Explain the basic components of soil. [U] / [CO4]
9. What is meant by solid waste management? [U] / [CO4]
10. What is meant by effluent? [K] / [CO5]
11. Name any two microorganisms used in effluent treatment. [U] / [CO5]
12. Explain the term phytoremediation. [U] / [CO5]

[10 x 2 = 20]

Part B
Short Answer Questions
Answer any 6 questions
Each question carries 5 marks

13. Briefly explain about solar energy and the applications of solar energy. [A] / [CO1]
14. Define environmental management system. [K] / [CO1]
15. Briefly explain the determination of Biochemical Oxygen Demand (BOD) of water sample. [U] / [CO2]
16. Discuss about water sampling techniques. [U] / [CO2]
17. Explain ozone layer depletion and describe its major consequences. [U] / [CO3]
18. Explain the Chernobyl disaster and discuss its major consequences. [U] / [CO3]

19. State and explain the three R's of waste management. [U] / [CO5]
20. Describe the different methods used for the management of toxic waste. [A] / [CO5]

[6 x 5 = 30]

Part C

Short Essay

Answer any 2 questions

Each question carries 10 marks

21. Describe briefly on the sources and biochemical effects of the pollution caused by PAN, cyanides(CN), Pesticides and carcinogenic substances. [U] / [CO1]
22. Explain the sources and effects of water pollution. [U] / [CO2]
23. Explain the analysis of following parameters in soil
    a) Nitrogen b) Phosphorous c) Calcium d) Magnesium [An] / [CO4]
24. Illustrate the methods used for wastewater treatment. [A] / [CO5]

[2 × 10 = 20]
